Examples:

Potash alum [$K_2SO_4.Al_2(SO_4)_3.24.H_2O$]; Sodium alum [$Na_2SO_4.Al_2(SO_4)_3.24.H_2O$] , Ammonium alum [$(NH_4)_2SO_4.Al_2(SO_4)_3.24.H_2O$], Chrome alum [$K_2SO_4.Cr_2(SO_4)_3.24.H_2O$].

Alums in general are more soluble in hot water than in cold water and in solutions they exhibit the properties of constituent ions.

Preparation:

The alunite the alum stone is the naturally occurring form and it is $K_2SO_4.Al_2(SO_4)_3.4Al(OH)_3$. When alum stone is treated with excess of sulphuric acid, the aluminium hydroxide is converted to aluminium sulphate. A calculated quality of potassium sulphate is added and the solution is crystallised to generate potash alum. It is purified by recrystallisation.

$$K_2SO_4.Al_2(SO_4)_3.4Al(OH)_3 + 6H_2SO_4 \longrightarrow K_2SO_4 + Al_2(SO_4)_3 + 12\ H_2O$$

$$K_2SO_4 + Al_2(SO_4)_3 + 24\ H_2O \longrightarrow K_2SO_4.Al_2(SO_4)_3.24\ H_2O$$

Properties

Potash alum is a white crystalline solid it is soluble in water and insoluble in alcohol. The aqueous solution is acidic due to the hydrolysis of aluminium sulphate it melts at 365 K on heating. At 475 K loses water of hydration and swells up. The swollen mass is known as burnt alum. Heating to red hot it decomposes into potassium sulphate, alumina and sulphur trioxide.

$$K_2SO_4.Al_2(SO_4)_3.24\ H_2O \xrightarrow{500\ K} K_2SO_4.Al_2(SO_4)_3 + 24\ H_2O$$

$$K_2SO_4.Al_2(SO_4)_3 \xrightarrow{\text{Red hot}} K_2SO_4 + Al_2O_3 + 3SO_3$$

Potash alum forms aluminium hydroxide when treated with ammonium hydroxide.

$$K_2SO_4.Al_2(SO_4)_3.24\ H_2O + 6NH_4OH \longrightarrow K_2SO_4 + 3(NH_4)_2SO_4 + 24\ H_2O + 3Al(OH)_3$$

Uses of Alum:

1. It is used for purification of water
2. It is also used for water proofing and textiles
3. It is used in dyeing, paper and leather tanning industries
4. It is employed as a styptic agent to arrest bleeding.

2.3 Group 14 (Carbon group) elements:

2.3.1 Occurrence:

Carbon is found in the native form as graphite. Coal, crude oil and carbonate rocks such as calcite, magnesite etc... contains large quantities of carbon in its combined form with other elements. Silicon occurs as silica (sand and quartz crystal). Silicate minerals and clay are other important sources for silicon.

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